

Building angular distribution for extreme observations using spectral clustering

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1 Introduction

The behavior of heavy-tailed multivariate random vectors above large radial thresholds has been extensively studied in various contexts. Detecting sparse patterns in rare events is particularly important for understanding multivariate extreme events Engelke and Ivanovs (2020). Modern methods for thresholding heavy-tailed distributions increasingly rely on machine learning techniques. For example, Drees and Sabourin (2019) employs principal component analysis to threshold observations radially.

In this report, we will focus on spectral clustering application to max-linear frameworks that were introduced in Davis and Resnick (1989). Other applications include the use of a k -means algorithm for the estimating the clusters Janßen and Wan (2019) or the use of Bayesian network Klüppelberg and Lauritzen (2019).

In the next section, we will discuss theoretical results of both multivariate extreme value analysis as well as guarantees provided by the spectral clustering. Then, we will provide numerical experiments based on spectral clustering on max-linear models.

2 Theoretical background

In this section, we describe the the main results provided by Medina et al. (2024), we will focus first on the multivariate extreme point of view. Then, we will introduce the notion of spectral clustering, before combining the two concepts in a framework for estimating the concentration clusters of the extreme distribution.

2.1 Angular distribution for multivariate distributions

2.1.1 RECALLS FROM MULTIVARIATE EXTREME VALUE THEORY

We first recall the results concerning regular variation for multivariate distributions. Given a random vector of dimension d , $\mathbf{X} = (X_1, \dots, X_d)$ and an exponent $\alpha > 0$, we say that $\mathbf{X}$ varies regularly, if the survival function of $\|\mathbf{X}\|$: $x \mapsto \mathbb{P}(\|\mathbf{X}\| > x) \in RV(\alpha)$ i.e. $\lim_{t \rightarrow \infty} \frac{\mathbb{P}(\|\mathbf{X}\| > tx)}{\mathbb{P}(\|\mathbf{X}\| > t)} = x^{-\alpha}$. This last limit characterizes the tail of the distribution of $\mathbf{X}$ in the multivariate context.

Furthermore, the paper focuses on determining the asymptotic angular distribution Γ on

the unit sphere $\mathbb{S}^{d-1}$ of $\mathbb{R}^d$:

$$\Gamma \leftarrow \mathbb{P} \left(\frac{\mathbf{X}}{\|\mathbf{X}\|} \in \cdot \mid \|\mathbf{X}\| > t \right), \text{ when } t \rightarrow \infty, \quad (1)$$

where $\leftarrow$ denotes the weak convergence towards an asymptotic distribution on the unit sphere.

2.1.2 LINEAR MODEL OF FACTORS

One of the core ideas of the paper is to build $\mathbf{X} \in \mathbb{R}^d$ using a linear model of different factors $\mathbf{Z} \in \mathbb{R}^p$, each factor being independent and identically distributed from the others, i.e. to build a non-negative matrix $A = (\mathbf{A}_1 \dots \mathbf{A}_p) \in \mathbb{R}^{d \times p}$, such that:

$$\mathbf{X} = A\mathbf{Z}. \quad (2)$$

Furthermore, we assume that each factor Z_k , with $k \in \{1, \dots, p\}$ follows an asymptotic Pareto distribution, that is to say $\mathbb{P}(Z_k > z) \sim_{z \rightarrow \infty} cz^{-\alpha}$, with $c, \alpha > 0$. The aim of the linear model is to identify the effect of extreme values for one or several factors in the vector $\mathbf{Z}$ over the angular distribution of $\mathbf{X}$. For instance, if we build a moving-average (MA) process $\mathbf{X}_t = \sum_{j=0}^{\tau} a_j \mathbf{Z}_{t-j}$, the factors of the model would be the random independent and identically distributed variables $\mathbf{Z}$, and the factor matrix is given by the parameters $(a_j)_{j=0}^{\tau}$ of MA process.

In this setting we can show that the angular distribution Γ is defined over atoms $\mathbf{a}_j := \frac{\mathbf{A}_j}{\|\mathbf{A}_j\|} \in \mathbb{S}^{d-1}$ with masses $\frac{\|\mathbf{A}_j\|^\alpha}{w}$ given by the j -th column of A , for $j \in \{1, \dots, p\}$ and $w = \sum_{j=1}^p \|\mathbf{A}_j\|^\alpha$ being a normalization constant:

$$\Gamma = \sum_{j=1}^p \frac{\|\mathbf{A}_j\|^\alpha}{w} \delta_{\mathbf{a}_j}.$$

The argument to use a clustering algorithm makes sense here as the asymptotic angular distributions concentrates on points. Another point to mention is that we do not have necessarily p distinct concentration points, some clusters may end up around the same concentration point, thus we can infer that there exist a number $p' < p$ of distinct clusters. Furthermore, determining the concentration points $\mathbf{a}_j$ for $j \in \{1, \dots, p\}$ is easy in theory if we know how to construct A , otherwise, we have to estimate $\hat{\mathbf{a}}_j$.

To quantify the convergence of the angular distribution towards the points $(\mathbf{a}_j)_{j=1}^p$, it can be shown that for a sequence of real numbers $(u_n)_{n \in \mathbb{N}}$ diverging towards $+\infty$, we have the weak convergence of:

$$\mathbb{P} \left(u_n \left(\frac{\mathbf{X}}{\|\mathbf{X}\|} - \mathbf{a}_k \right) \in \cdot \mid \|\mathbf{X}\| > u_n, Z_k > u_n/w^{\alpha-1} \right) \xrightarrow{n \rightarrow \infty} \mathbb{P} \left(\frac{1}{\|\mathbf{A}_k\|^2 W_\alpha} (S_{1,k}, \dots, S_{d,k})^T \in \cdot \right),$$

where $S_{j,k} := \sum_{i=1}^d \left(\mathbf{A}_j^{(i)} \right)^2 \left(\mathbf{A}_{-j}^{(k)} \mathbf{Z} \right) - \mathbf{A}_j^{(k)} \mathbf{A}_j^{(i)} \left(\mathbf{A}_{-j}^{(i)} \mathbf{Z} \right)$, with $\mathbf{A}_{-j}^{(i)} \mathbf{Z} := \sum_{\substack{m=1 \\ m \neq j}}^p \mathbf{A}_m^{(i)} Z_m$ and W_α

follows a Pareto distribution of parameter α independent from each of the factors. This

weak convergence is really interesting as it ensures that the empirical angular distribution will be distributed around the atoms $(\mathbf{a}_j)_{j=1}^p$.

We note that the conditional distribution depends on one specific threshold exceeding factor. Indeed, to build the graph associated with Γ , we will have to isolate extreme values achieved by the different factors. To ensure the connection between the exceeding of u_n by $\|\mathbf{X}\|$ and "extreme values" from only one of the factors, we can actually control the growth rate of $(u_n)_{n \in \mathbb{N}}$ and build a sequence $(h_n)_{n \in \mathbb{N}}$ that depends on $(u_n)_{n \in \mathbb{N}}$ such that when $\|\mathbf{X}\| > u_n$, then there is at most one Z_j , for $j \in \{1, \dots, p\}$, such that $Z_j > h_n$.

In fact, we can deduce from a sample $(\mathbf{Z}^{(1)}, \dots, \mathbf{Z}^{(n)})$ of size n , that the sequence of events:

$$B_n = \left\{ \forall i \in \{1, \dots, n\} \text{ such that } \|\mathbf{X}^{(i)}\| > u_n, \exists! j \in \{1, \dots, p\}, Z_j^{(i)} > h_n \right\},$$

is almost sure when $n \rightarrow \infty$. Using this previous result we can quantify the distance between the concentration points $(\mathbf{a}_j)_{j=1}^p$ and the empirical distribution. For $\delta > 0$, we can establish the upper bound for n large enough such that $\mathbb{P}(B_n) > 1 - \delta$:

$$\left\| \frac{\mathbf{X}^{(i)}}{\|\mathbf{X}^{(i)}\|} - \mathbf{a}_j \right\| \leq \frac{8(a^*)^2 h_n}{\|\mathbf{A}_j\|^\alpha u_n}, \quad j \in \{1, \dots, p\}, \quad (3)$$

for i such that $\|\mathbf{X}^{(i)}\| > u_n$ and $Z_j^{(i)} > \frac{u_n}{a^*}$ where $a^* := \sqrt{d} \max_{\substack{1 \leq i \leq d \\ 1 \leq j \leq p}} \mathbf{A}_j^{(i)}$. The goodness of

the upper bound depends on the sequences $(u_n)_{n \in \mathbb{N}}$, $(h_n)_{n \in \mathbb{N}}$ and the sample size n . The paper gives an example of sequence $(h_n)_{n \in \mathbb{N}}$, for $n \in \mathbb{N}$, $h_n = u_n^{(\alpha-1)/4} n^{(2-\alpha)/(4\alpha)}$. Furthermore, this result is interesting as we can compute a deterministic bound for the distance between an angular sample and the center of the cluster it belongs to. This will be useful, in the context of the k -nn graph clustering.

In the next sections, we will consider the sets $I_{n,j} = \left\{ i \in \{1, \dots, n\}, \|\mathbf{X}^{(i)}\| > u_n, Z_j^{(i)} > h_n \right\}$ and their cardinals $N_{n,j} := |I_{n,j}|$, as well as $N_n := \sum_{j=1}^p N_{n,j}$.

2.2 Spectral clustering

The authors chose to leverage the method of spectral clustering on the angular samples to explore the dependence of the multivariate extremes distribution.

The spectral clustering is a clustering method which consists in using the eigenvectors of the Laplacian matrix $\mathcal{L}$ of a well-defined graph $\mathcal{G}$ to determine relevant clusters. We will present this below.

2.2.1 DEFINITION OF A RELEVANT GRAPH

Let $\mathbf{X}^{(1)}, \dots, \mathbf{X}^{(n)}$ be the i.i.d. multivariate samples, and u_n a threshold. Only the samples with a norm higher than u_n are considered to be extreme. Let $\mathcal{V}_n$ the indexes of samples that are above the threshold.

To formulate a relevant similarity between two multivariate extreme vectors, the authors propose two ways of defining a graph with nodes in $\mathcal{V}_n$:

- At first, the authors define an ϵ -neighbourhood graph. For each couple of extreme samples $(\mathbf{X}^{(i)}, \mathbf{X}^{(j)})$, $i, j \in \mathcal{V}_n$, there is an edge between them if and only if

$$\rho \left(\frac{\mathbf{X}^{(i)}}{\|\mathbf{X}^{(i)}\|}, \frac{\mathbf{X}^{(j)}}{\|\mathbf{X}^{(j)}\|} \right) < \epsilon$$

where ρ is a distance on $\mathbb{S}_{d-1}$, usually the Euclidean distance is chosen. In this case, the adjacency matrix is $\mathbf{W}$ with weights 1 for ϵ -neighbours and 0 else.

- Then, the authors focus on a k -nearest-neighbours graph - k -NN graph – which connects two nodes i and j if i is one of the k -NN of j or if j is one of i 's given a specific metric.

In this case, the associated weight is:

$$\mathbf{W}_{i,j} = \begin{cases} d \left(\frac{\mathbf{X}^{(i)}}{\|\mathbf{X}^{(i)}\|}, \frac{\mathbf{X}^{(j)}}{\|\mathbf{X}^{(j)}\|} \right), & \text{if } i \text{ and } j \text{ are connected,} \\ 0, & \text{else.} \end{cases}$$

where d is a universal kernel, $d(x, y) = \exp(-\|x - y\|)$ for instance.

2.2.2 THE SPECTRAL CLUSTERING ALGORITHM

For a given graph with weighted adjacency matrix $\mathbf{W} \in \mathbb{R}^{n \times n}$, we define $\mathbf{D} = \text{diag}((d_i)_{i \in \mathcal{V}_n})$, where $d_i := \sum_{j \in \mathcal{V}_n} \mathbf{W}_{i,j}$ is the degree of node i . We can then define the symmetrical Laplacian matrix of the graph:

$$\mathbf{L} = \mathbf{I} - \mathbf{D}^{-\frac{1}{2}} \mathbf{W} \mathbf{D}^{-\frac{1}{2}}.$$

The spectral algorithm which uses the previously defined symmetrical Laplacian is defined in Ng et al. (2002) as follows:

1. We consider the $p' \in \mathbb{N}^*$ smallest eigenvalues $\lambda_1, \dots, \lambda_{p'}$ of $\mathbf{L}$ and the associated eigenvectors $U_1, \dots, U_{p'}$.
2. We define $\mathbf{U} \in \mathbb{R}^{n \times p'}$, such as the i th column of $\mathbf{U}$ corresponds to U_i .
3. We then obtain a matrix $\mathbf{V}$ by normalizing the rows of $\mathbf{U}$.
4. We can cluster the normalized rows using a K-means clustering (with p' clusters).
5. We then identify the i th node as the i th row, and thus obtain p' clusters for the nodes.

The algorithm is useful for what the paper accomplishes. We can define the connected components of the graph i.e. the subsets $A \subset \mathcal{V}_n$ such that for all $i, j \in A$, there is an edge between i and j , and for all $i \in A$ and $j \in \mathcal{V}_n \setminus A$, there is no edges between i and j . In fact, the multiplicity of the eigenvalue equal to 0 of the graph Laplacian $\mathbf{L}$ is equal to the number of connected components $A_1, \dots, A_{p'}$ of the graph. Furthermore, the eigenspace of 0 is generated by the vector space of indicator vectors $\text{Vect}(\mathbb{1}_{A_1}, \dots, \mathbb{1}_{A_{p'}})$ as stated in Luxburg (2007). Using the spectral clustering algorithm with p' components, we can identify the connected component of the graph.

2.3 Application of spectral clustering to the extreme linear factor model

For simplicity sake, we make more assumptions over the distributions to ensure that we have statistical guarantees over the spectral clustering outcome. We will notably focus on the case $d = 2$.

We make further assumptions on the marginal distributions of the factors of $\mathbf{Z}$, that is to say that the marginal density function f_Z is positively lower bounded on every compact intervals and upper bounded. We also impose that for all z , $\frac{z^{-(1+\alpha)}}{B} \leq f_Z(z) \leq Bz^{-(1+\alpha)}$ for $B > 0$. Let

$$T_{n,j} := \frac{\mathbf{A}_{-j}^{(3-j)} \mathbf{Z}}{w_j^2 Z_j / u_n},$$

with $j \in \{1, 2\}$. We can find that the density function of $T_{n,j}$ is upper bounded on $\mathbb{R}$ by $G \in \mathbb{R}_+^*$.

For a fixed $j \in \{1, 2\}$, with an appropriate sequence $k_n = o_{n \rightarrow \infty}(nu_n^{-\alpha})$, if $k_n > G \log(n)$, then the spectral clustering algorithm will consistently distinguish all the clusters of extremes that come from the linear model in (2). This sequence k_n will be used as the number of neighbors in the the extreme k_n -NN graph. With this in mind, we can actually infer that the number of connected components in the k_n -NN graph will be the same as the underlying clusters of the linear model.

In practice, the hypothesis over G can be relaxed, by choosing an arbitrary $G_0 > 0$ such that $k_n > G_0 \log(N_n)$ and $k_n = o_{n \rightarrow \infty}(N_n)$. One practical solution for such sequence is the sequence whose general term is $k_n = \left\lceil \frac{N_n}{\tau \log(N_n)} \right\rceil + 1$, for all $n \in \mathbb{N}^*$ and some $\tau > 1$.

We can now estimate the concentration points $\hat{\mathbf{a}}_j$, for $j \in \{1, \dots, p'\}$. Knowing the p' clusters outputted by the spectral clustering algorithm, one can easily compute $\hat{N}_{n,j}$ by counting how many samples fall in the j th cluster $\hat{I}_{n,j}$. We can deduce as well for $j \in \{1, \dots, p'\}$:

$$\begin{aligned} \hat{\pi}_j &= \frac{\hat{N}_{n,j}}{N_n}, \\ \hat{\mathbf{a}}_j &= \frac{1}{\hat{N}_{n,j}} \sum_{i \in \hat{I}_{n,j}} \frac{X_i}{\|X_i\|}. \end{aligned}$$

The first line corresponds to the empirical weights $\hat{\pi}_j$ of the j th cluster. We can also show the convergence of probability of the empirical weights $\hat{\pi}_j \xrightarrow[n \rightarrow \infty]{P} \frac{\|\mathbf{A}_j\|^\alpha}{w}$, as well as $\hat{\mathbf{a}}_j \xrightarrow[n \rightarrow \infty]{P} \mathbf{a}_j$ for all distinct cluster $j \in \{1, \dots, p'\}$.

3 Numerical experiments

3.1 Linear Factor Model without noise

First of all, we apply the clustering algorithm on generated data that follows the Linear Factor Model.

For that, we consider a factor matrix $A \in \mathbb{R}^{d \times p}$, where $d, p \in \mathbb{N}^*$ and we generate $\mathbf{X} = A\mathbf{Z}$, where $\mathbf{Z}$ is a multivariate variable of size p , each coordinate following a Fréchet law of parameter $-\frac{1}{\alpha}$, which means $\exists c, \forall k, \mathbb{P}(Z_k > z) \sim_{z \rightarrow \infty} cz^{-\alpha}$.

For the sampling of the $\mathbf{X}$ realization, we consider i.i.d. samples $\mathbf{Z}$ of p independent Fréchet variables.

We only consider here $N_n = 10^3$ extreme values, $\alpha = 1.3$, $p = 3$ and $d = 2$ in various simulations, where N takes for values 10^5 and 10^7 . The factor matrix we will use is

$$A = \begin{bmatrix} 0.2 & 0.5 & 1 \\ 1 & 0.5 & 0.2 \end{bmatrix}.$$

We define the threshold is implicitly chosen in order to select N_n samples, and for the adjacency matrix we use a k -nn algorithm with $k = \lceil \frac{N_n}{1.5 \log(N_n)} \rceil + 1$ as well as the kernel operator $e^{-\|x-y\|}$. In the Figure 1 are displayed the N_n extremes for n sample of the model, the results of the spectral clustering and both the empirical mean position for each cluster compared to the associated $\frac{\mathbf{A}_j}{\|\mathbf{A}_j\|}$, as well as the comparison between the empirical weight associated to each cluster and the theoretical limit weight $\frac{\|\mathbf{A}_j\|^\alpha}{\sum_i \|\mathbf{A}_i\|^\alpha}$.

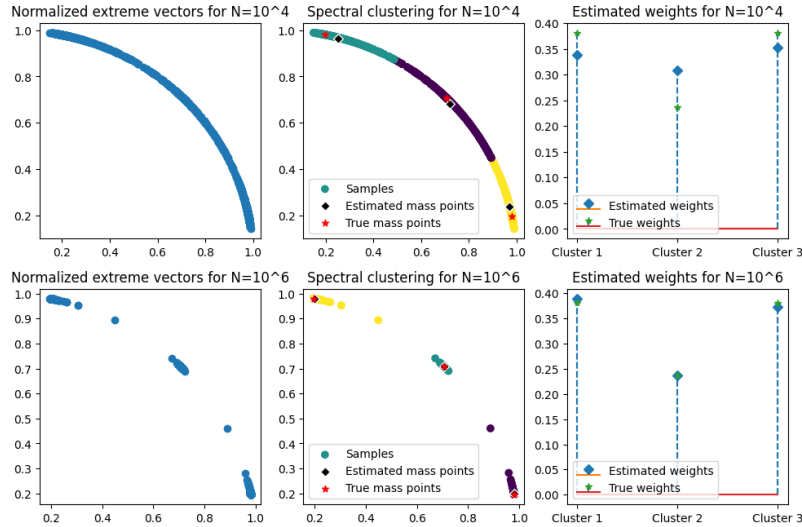


Figure 1: Results of the spectral clustering algorithm for different sample sizes N . On the left, the extreme angular distribution (defined over a threshold). On the middle, the result of the spectral clustering on the angular distribution. On the right, stem plot of estimated and true weights.

As for the robustness of these results, on Figure 2, we display the boxplots of the Euclidean distance between the aggregate of the mean positions for each cluster and their theoretical limit for $n = 100$ experiences, as well as the boxplot of the empirical weights and their limit for n implementations.

These numerical experiments seem to be consistent with the theoretical predictions regarding the relevance of the algorithm and its convergence given a specific set of parameters. With other parameters, the convergence might be slower – for instance if α is higher or if two angles of A are arbitrary close – or faster.

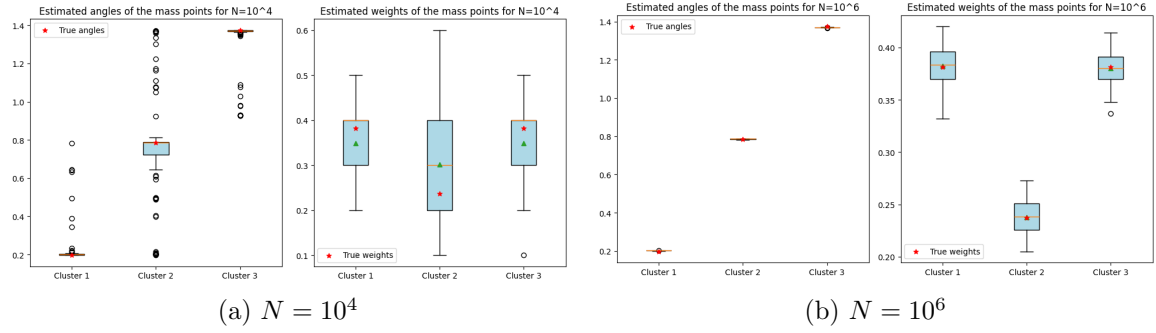


Figure 2: Box plot of the extreme distribution estimations for both mass points and weights for different sample sizes N .

3.2 Another toy example

It seems that for data that would not follow a LFM, some strong properties should be necessary for the algorithm to give relevant results ; for instance the coercivity of the function $\mathbf{Z} \rightarrow \mathbf{X}(\mathbf{Z})$, and the convergence of $\frac{\mathbf{X}(t_1, \dots, t_{i-1}, t, t_{i+1}, \dots, t_d)}{\|\mathbf{X}(t_1, \dots, t_{i-1}, t, t_{i+1}, \dots, t_d)\|}$ when $t \rightarrow \infty$ and $\frac{t_1}{t}, \dots, \frac{t_{i-1}}{t}, \frac{t_{i+1}}{t}, \dots, \frac{t_d}{t} \rightarrow 0$ with $1 \leq i \leq d$. Let us consider for instance the same previous model with the same A , and same parameters, but considering $\mathbf{X}' = (\sqrt{|X_1|}, \dots, \sqrt{|X_p|})^T$. As displayed on a specific sampling in Figure 3, the clustering algorithm seems to converge similarly to the LFM model, the mass points being concentrated on the normalized square-rooted columns of A . This could be expected since the data is generated in a very similar way, just using the square-root function which is coercive and monotone.

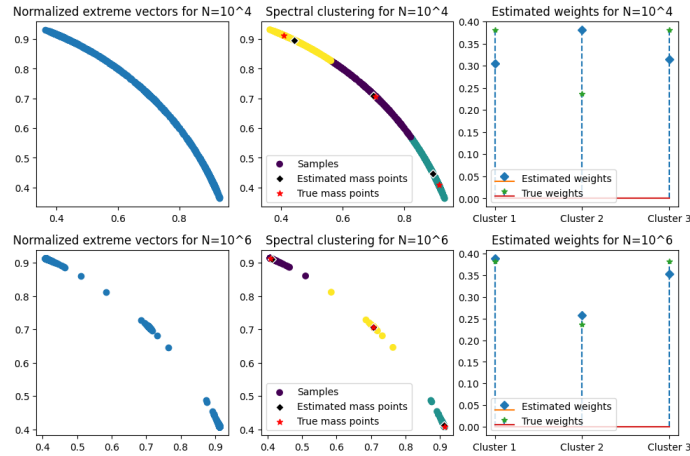


Figure 3: Results of the spectral clustering algorithm for different sample sizes N . On the left, the extreme angular distribution (defined over a threshold). On the middle, the result of the spectral clustering on the angular distribution. On the right, stem plot of estimated and true weights.

3.3 Rain Data in South England

We fit a moving average (MA) process of order 3 onto a dataset of daily precipitation levels in south England. Thus, we consider the factors $\mathbf{Z} = (Z_t, Z_{t-1}, Z_{t-2}, Z_{t-3})$ and the output of the linear model $\mathbf{X} = (X_t, X_{t-1})$. The intuition behind this model is to consider that precipitations are not independently distributed and in fact the auto-correlation factor for some lag $h > 0$ is not null. Following the Figure 4, which were obtained by thresholding on X_t , the results are the following, globally the estimated masses and weights are adequate. The angular distributions for each cluster is not as concentrated around each mass points as in the examples, the upper bound for the Euclidean distance between the observations and the mass point as given in Equation 3 is still valid as the value of the upper bound is around 1.30.

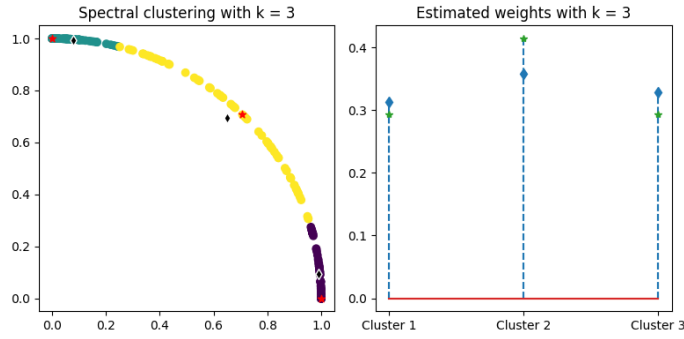


Figure 4: Results of the spectral clustering algorithm for the rain dataset.

4 Conclusion

This paper offers an robust clustering algorithm for the angles of extreme multivariate vectors generated according a linear factor model and underlying asymptotic Pareto distributions. The algorithm consists in applying spectral clustering on a k -nn graph built with a kernel, and associated to the angles of the extreme multivariate vectors above a specific threshold. The robustness results concerns the limit angular distribution converging to a weighted sum of Dirac masses, each one associated to a specific factor in the non-degenerate case.

Besides the theoretical examples, this algorithm seems to apply well on real world data when there are few underlying factors that cause the studied variables to take extreme values. The underlying distribution of the cause does not even need to be heavy-tailed, since we can artificially transform the marginal distributions following Janßen and Wan (2020).

Furthermore, as the paper mentioned the algorithm also seems to converge well when the data is not exactly generated by a LFM. This is also promising for the application on real world data.

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